

Selection Structures

if statements mostly

Selection structures

Almost every puzzle can be solved with a set of `if` statements



If Statements

```
1  x = int(input("Please enter an integer: "))
2
3  if x < 0:
4      x = 0
5      print('Negative changed to zero')
6  elif x == 0:
7      print('Zero')
8  elif x == 1:
9      print('Single')
10 else:
11     print('More')
```

There can be **zero or more** elif statements, and **zero or one** else statements

If Statements

```
1  <if> <condition> <colon>
2  <tab> <body>
3  <tab> <body>
4  <elif*> <condition> <colon>
5  <tab> <body>
6  <tab> <body>
7  <else> <colon>
8  <tab> <body>
```

Nested decisions

An `if` statement can contain another `if` statement inside its body — this lets you check a second condition, but only once the first one is true

```
1  <if> <condition_1> <colon>
2    <tab> <if> <condition_2> <colon>
3      <tab> <tab> <body>
4      <tab> <tab> <body>
5      <tab> <else> <colon>
6      <tab> <tab> <body>
7    <else> <colon>
8  <tab> <body>
```

Each level of nesting adds another `<tab>` of indentation

Nested decisions

```
1 age = int(input("Enter your age: "))
2 has_ticket = input("Do you have a ticket? (y/n): ") == "y"
3
4 if age ≥ 18:
5     if has_ticket:
6         print("Welcome in!")
7     else:
8         print("You need a ticket")
9 else:
10    print("Sorry, you must be 18 or older")
```

The inner `if / else` only runs when `age ≥ 18` is true

Nested vs. combined conditions

These two are equivalent, nesting can often be flattened with `and`

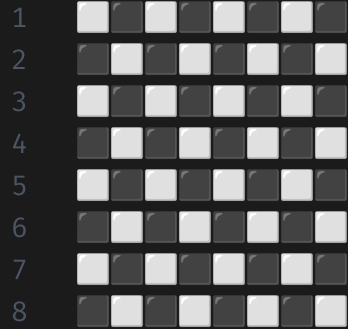
```
1  # nested
2  if age ≥ 18:
3      if has_ticket:
4          print("Welcome in!")
```

```
1  # combined
2  if age ≥ 18 and has_ticket:
3      print("Welcome in!")
```

Use nesting when the inner check only makes sense **after** the outer one, especially if each branch needs different follow-up logic

Practice problem

Given a square's (row, col) on the chessboard below, determine if it's light or dark:



Practice problem

Given the tic-tac-toe board below, has anyone won?

1	x		o		x
2					
3	o		x		o
4					
5	x		.		.

Practice problem

The player * is trying to move right. Will they hit the wall # ?

```
1  . . . . .
2  . # # . .
3  . . * . .
4  . . . . .
```

Practice problem

A user enters a number. Print whether it's even or odd.

Practice problem

Given three numbers, print the largest one.

Practice problem

Given a year, determine if it's a leap year.

Practice problem

Given three side lengths, determine if they form a valid triangle. (The sum of any two sides must be greater than the third.)

Practice problem

Given a password, check if it's "strong": at least 8 characters, contains a digit, and contains a symbol.

Practice problem (nested)

A user enters an age and whether they have a ticket. Only check for a ticket if they're old enough (18+); otherwise refuse entry right away.

Practice problem (nested)

Given a number, print "positive and even", "positive and odd", or "not positive", only check even/odd once you know it's positive.

Practice problem (nested)

Given a username and password, log the user in only if the username exists **and** the password matches. Print a different message for "unknown username" vs. "wrong password".

Practice problem (nested)

Given a temperature and whether it's raining, decide what to wear: only worry about a raincoat if it's cold **and** raining.